MILITARY SPECIFICATION

TURRET SUBSYSTEM, UNIVERSAL (UTS), FOR ARMAMENT SUBSYSTEM, HELICOPTER, 20MM: M197El and M97E2, TURRET CONTROL-UNIT FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for-use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the turret control unit (TCU).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, Standards, and Handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herin.

SPECIFICATIONS

MILITARY

MIL-B-5087	-	Bonding, Electrical and Lightning Protection, for Aerospace Systems
MIL-E-5400	-	Electrical Equipment, Airborne, General Specification for
MIL-W-63150	-	Weapons and Support Material, Standard Quality Assurance Provisions for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT

11830908	-	Turret Control Assembly

PACKAGING

P11830908	-	Turret Control Assembly

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117	-	Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specified acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Item definition. The turret control unit, Drawing 11830908, shall be configured for use in an AH-IS helicopter equipped with the XM97E1 20mm Armament Subsystem. This control is a totally enclosed box which contains the required circuitry to translate signals from the aircraft turret electrical system into appropriate signals for turret drive motor control. The following functions shall be performed by the turret control unit (TCU)

a. Convert resolver type signals into appropriate split field series wound d-c drive motor excitation.

b. Provide a d-c analog signal indicating the amplitude of the input position command signal (linear up to 750mV rms on the input).

c. Provide an electrical interface to control emergency stow initiation.

d. Provide mode selection as to which position comnand signal controls the turrent drive motor.

e. Provide boresight adjustment capability in the fixed forward mode.

f. Provide a logic signal indicating gunner stow position is commanded.

g. Provide an action signal interlock when fixed forward signal is selected.

h. Provide slow slew rates for azimuth and elevation turret responses.

i. Provide for the constant gain transfer function in the azimuth servo loop as a function of the Cos El signal.

j Provide filtered zener power for the limit switch assemblies.

3.1.1 Major component list. The major component list shall be as specified in Table I.

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TABLE I. Major component list

Component	Ref Des	Drawing Number
Case, Electrical Equipment	-	189F81 9
Component Assembly, Demodulator, Power Supply	Al	189F960
Component Assembly, Filter, Limit Switch	A2	11830909
Component Assembly, Current Loop	A3 & A4	11830800
Component Assembly, Interconnection	A5	J 1830799
Component Assembly, Driver		139C7957
Component Assembly	A8	132D1354

3.1.2 Power supply. The TCU Shall meet the requirementS Of MIL-STD-704. During normal operation of the primary electrical system, the TCU shall meet the requirements of Paragraph 5.3.2 of MIL-STD-704. During abnormal operation of the electrical system, the TCU shall meet the requirements of Paragraph 5.303 of MIL-STD-704. The power supply for all acceptance tests shall be As specified in Table II.

Table II. Electrical Power

Pin	Power Form	Nominal	Peak
J2-G J2-E	28 Vdc Low Level Return for J2-G	1.0 ampere	2 amperes
E1 E2	28 Vdc High Level Return for El	10 amperes	200 amperes
J2-H J2-J	115 v, 400 Hz, 1 0 Return for J2-H	70 volt-amperes	95 volt-amperes
J2-L J2-K	26 V, 400 Hz, 1 0 Return fOr J2-L	1 volt-ampere	2 volt-amperes

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3.1.2.1 Ground. In order to facilitate installation in the helicopter, the equipment enclosures and mounting provisions shall be designed to allow electrical bonding in accordance with MIL-B-5087.

3.2 Material. Materials shall be in accordance with applicable drawings, specifications, and standards.

3.3 Assembly. The turret control assembly shall comply with all requirements specified on Drawing (Dwg) 11830908, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.4 Interchangeability. All parts having the same part number shall be completely interchangeable with each other and with like parts of other manufacture with respect to installation and performance. Parts joined together by welding, brazing or riveting shall be interchangeable-as assemblies. Interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.5 First article inspection. This specification contains technical provisions for first article inspections. Requirements for the submission of first article inspection by the contractor shall be as specified in the contract and 4.3.

3.6 Electronic performance characteristics.

3.6.1 Performance. The turret control unit shall have the resultant characteristics of the major subassemblies as referenced in Table I. The interconnecting cabling between these assemblies is defined on Drawing 11830939.

3.6.1.1 Electronic characteristics. The TCU electronic characteristics shall be as described in Table III. Each connector pin is defined with a function and a characteristic. Where it is stated that the signal is not used, no external connection is to be made or measurement taken.

3.6.1.2 Input/output relationships. The input/output relationships shall be as specified in the following paragraphs. Unless otherwise specified, these relationships shall be with power applied in accordance with 3.1.3 and Table II.

3.6.1 .2.1 Azimuth preamplifier and demodulator-Action Mode. With 26.5 (+5) Vdc action relay control applied to Pin J1-K and Zero Vdc fixed forward mode select voltage on Pin J2-U, the azimuth demodulator output dc voltage on Pin J2-D shall be a function of the 400 Hz COS EL input signal voltages at J3-J as tabulated in Table IV for sight Az error input voltages J1-M up to 0.5 Vrms based on the average of both polarities.

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TABLE III. Electronic characteristics

Connector	Function	Characteristics
j1-A	Shield	Chassis ground
J1-B	Sight Elevation Error Signal Input	0-12 Vrms, 400 Hz +5% sinusoidal, to an input impedance of 0.047 F in parallel with 400K ohms and the input impedance of a solid state op-amp.
JI -C	Return forJ1-B	
J1-D	Not used	
J1 -E	Signal, Ground	
JI-F	Not Used	
J1 -G	Not Used	
J1 -H	Not Used	
J1-J	Not Used	
J1-K	Action Relay Control Voltage Input	26.5 (+5 Vdc) will enable Az and EL error signal circuits with input at J1-B and J1-M.
J1 -L	Not Used	
J1 -M	Sight Azimuth Error Signal Input	0-12 Vrms, 400 Hz +5% sinusoidal, input impedance of 0.047 F in parallel with 400K ohms and the input impedance of a solid state op-amp.
Jl -T	Return for J1-M	
J1-N	Sin Az Cos EL Signal Output	Parallel with J3-G, 0-12 Vrms, 400 (+20)Hz sinusoidal
J1-P	Sin Az Cos EL Signal Return to Logic ControT Unit	Parallel with J3-F
J1 -R	Sin EL Signal Output “	Parallel with J3-D, 0-12 Vrms, 400 (+20)Hz sinusoidal.
J1-S	Sin EL Signal Return Logic Control Unit	Parallel with J3-E
J1 -U	Not Used	
J2-A	Not Used	

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TABLE III. Electronic characteristics (continued)

Connector	Function	Characteristics
J2-B	Not USed	
J2-C	Not Used	
J2-D	Azimuth Demodulator Output	0 to 5 Vdc proportional to magnitude and phase of the 400 Hz signals on J1-M and J3-J .
J2-E	28 V Low Level d-c, Return	Common return bus.
J2-F	Not used	
J2-G	28 V Low Level d-c Input Power	28 Vdc per MIL-STD-704, category B. Input current not to exceed 2.0 amperes.
J2-H	115 Vac 400 HZ power	115 Vac, 400 Hz per MIL-STD-704 Category B. Input a-c power 95 VA maximum ungrounded
J2-J	115 Vac 400 Hz Power Return	
J2-K	26 Vac 400 Hz Return	Reference signal for demodulator. (Must be same phase as system resolver signal excitation.)
J2-L	26 Vac 400 Hz Power	
J2-M	Not used	
J2-N	Elevation error gate and gunner Stow output	A normally open set of relay contacts is in series with J2-N output. The relay will be energized when J3-N is low and J1 -K or J2-U is at 28 VdC. Output, open circuited, is either 28 Vdc (low level) or 0.5 Vdc maximum.
J2-P	Elevation demodulator output	O to 5 Vdc proportional to magnitude and phase of the 400 Hz signal on J1-B.
J2-R	Not used	

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TABLE III. Electronic character istics (continued)

Connector	Function	Characteristics
J2-S	Slow Slew Limit Signal Output	With output voltage present, the impedance to ground is less than 100 ohms. Impedance will be greater than 100K ohms when the output is off.
J2-T	Shield Return	Chassis ground.
J2-U	Fixed Forward Mode Select Input	26.5 (+5) Vdc will enable the fixed forward mode.
J3-A	Left Limit Input Signal	Normally 28 Vdc when not at limit (low level). Limit state will be 0 (± 0.5) Wk.
J3-B	Right Limit Input Signal	Same input characteristics as J3-A.
J3-C	Down Limit. Input Signal	Same input characteristics as J3-A.
J3-K	Up Limit Input Signal	Same input characteristics as J3-A.
J3-D	Stn EL Input	(Parallel with JI-R) 0-12 Vrms, 400 Hz +5% sinusoidal, input impedance of 18K ohms minimum in parallel with 0.047 F through a normally closed relay contact (Open when 28 Vdc on J1-K).
J3-E	Sin EL Return to Turret	Low Level Ground (Parallel with J1-S).
J3-F	Sin Az Cos EL Input Return to the Turret	Low Level Ground bus.
J3-G	Sin Az Cos EL Input	Parallel with J1-N. 0-12 Vrms, 400 Hz +5% sinusoidal, input impedance of 18K ohms minimum in parallel with a 0.047 F through a normally closed relay contact (open when 28 Vdc on J1-K).

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TABLE III. Electronic characteristics (continued)

Connector	Function	Characteristics
J3-H	Cos EL Return	Low Level Ground bus.
J3-J	Cos EL Input	0-12 Vrms 400 Hz +5% sinusoidal, input impedance of one megohm minimum.
J3-L	Shield	Chassis ground.
J3-M	Limit Switch Ground Return	Low Level Ground bus.
J3-N	Emergency Stow Action Initiate Input	Normally +2.5 to 4.0 Vdc above ground (Diode isolated with anode at J3-N).
J3-R	Azimuth Tachometer Input	O to +90 Vdc from d-c generator.
J3-P	Return for J3-R	Ground bus.
J3-T	Elevation Tachometer Input	O to +90 Vdc from d-c generator.
J3-S	Return for J3-T	Ground bus
J3-U	Limit Switch Power SUpplY output	Connection to 28 Vdc (low level) a series with 27 (+10%) ohm 1 watt resistor and a 30 volt (+5%) zener diode shall be across-the output. (J3-U to J2-E)
J4-A	Turret Down Drive (when enabled) by an error signal at J1-B and 28 Vdc at J1-K)	Pulses of 28 Vdc (high level). Peak amplitude of varying width at a repetitive rate of 600 Hz. Output load (split series motor) shall be 0.1 ohms minimum in series with 150 H + 20% which saturates at 30 A. Each output shall have a coasting diode for back EMF protection.
J4-C	Turret Up Drive (when enabled)	Same characteristics as J4-A.

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TABLE III. Electronic characteristics (continued)

Connector	Function	Characteristics
J4-D	Turret Left Drive (when enabled by an error signal atJ1-M and 28 Vdc at J1-K)	Same characteristics as J4-A.
J4-F	Right Drive (when enabled)	Same characteristics as J4-A.
J4-E	Azimuth Motor Return	Continuity to stud terminal E2 via a suitable LC noise filter network including back EMF diodes.
J4-B	Elevation Motor Return	
J4-G	Motor Brake Supply	Continuity to stud terminal El (28 Vdc high level) via a suitable LC filter network.
E1	H.L. 28 Vdc Input Power	200 ampere maximum requirement (load dependent). Input to a LC noise filter network.
E2	H.L. 28 Vdc Power Return	Connected to the negative polarity leg of LC filter network.

TABLE IV. Gain values

Input Pin J3-J Vrms	Gain (Pin J2-D voltage level divided by Pin J1-M voltage level) Units-Vdc/Vrms
10.0	1.83 +15%
8.5	1.94±15%
6.4	2.60±15%

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The polarity of the output voltage shall be negative when theJ1-M input voltage is in phase with the 26 Vac voltage on Pin J2-L. The maximum ripple shall be 250 mV peak to peak on the voltage output at Pin J2-D with 1.0 Vrms input on Pin J1-M. A filter with a time constant of 4.3ms (+20 percent) shall be included in the test set to provide smoothing of the-voltage measured at Pin J2-D. With an input of 10.0 Vrms at Pin J3-J and an input of 5.0 Vrms or more at Pin J1-M, the output at Pin J2-D shall be 3.5 (+1.5) Vdc or -3.5 (+1.5) Vdc.

3.6.1 .2.2 Elevation Preamplifier and Demodulator-Action Mode. With 26.5 (+5) Vdc applied to the action relay control voltage input, Pin J1-K, and zero Vdc on the fixed forward mode select input, Pin J2-U, the elevation demodulator ouput voltage, at Pin J2-P, shall be a function of the signal elevation error 400 HZ signal voltage between dc PinJ1-B and PinJ1-C (signal return). The slope of the gain function relating voltage input at Pin J1-B to the dc output at J2-P shall be 1.7 Vdc per Vrms up to .7 Vrms. The maximum ripple shall be 250mV peak to peak on the voltage output at Pin J2-P with an input of 1.0 Vrms at J1-B. A filter with a time constant of 4.0 ms (+20 percent) shall be included in the test set to provide smoothing of the voltage measured at Pin J2-P. The polarity of the output voltage shall be positive when the error input signal is in phase with the 400 Hz voltage on Pin J2-L. For input voltages at Pin J1-B of 5.0 Vrms or more, the voltage output at Pin J2-P shall be 3.5 (±1.5) Vdc or-3.5 (±1.5) Vdc.

3.6.1 .2.3 Azimuth preaplifier and demodulator-Fixed Mode/Adjustments. With 26.5 (+5)Vdc applied to Pin J2-U and J1-k, the demodulator output voltages on_Pin J2-D shall be a function of the Sin Az COS EL input voltage input on Pin J3-G and the setting of the azimuth stow adjustment potentiometer (accessible through a hole in the assembly cover). The gain characteristics shall be modified by input voltage levels on Pin J3-J with the same gain values specified in Table IV. The stow potentiometer shall have an adjustment range such that the output at Pin J2-D can be varied from a minimum of +0.8 Vdc to a maximum of +1.5 Vdc when the voltage at Pin J3-G is zero Vrms and the voltage at Pin J3-J is 10 Vrms.

3.6.1 .2.4 Elevation preamplifier and demodulator-Fixed Mode/Adjustments. With 26.5 (+5)Vdc applied in Pin J2-U and Pin J1-K, the dc VOltage output at Pin J2-P shall be a function of the sin EL as voltage input at Pin J3-D and the setting of the elevation stow potentiometer (accessible through hole in the assembly cover). The slope of the gain function relating voltage at Pin J3-D and Pin J2-P shall be 1.7 Vdc per Vrms for ac input voltages up to 0.7 V based on the average at J2-P of both polarities, polarity depending on whether the input voltage at J3-D is in phase or 180 degrees out of phase with the power voltage at J2-L. The elevation stow potentiometer shall have an adjustment range from minimum of ± 0.7 Vdc to a maximum of ±1.3 Vdc at Pin J2-P when the input at J3-D is zero.

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3.6.1 .2.5 Normal stow mode. With all mode switching voltages at zero Vdc (Pin J2-U and Pin J1-K), the azimuth demodulator output voltage at Pin J2-D shall be as specified in 3.6.1 .2.3 with regard to the azimuth-stow adjustment. The elevation demodulator output voltage at Pin J2-P shall be zero Vdc when the input voltage at Pin J3-D is 1.92 Vrms (±15 percent) and 180 degrees out of phase with the voltage on Pin J2-L.

3.6.1.2.6 Output drive and control.

3.6.1 .2.6.1 Azimuth output drive and control. With 26.5 (±5) Vdc applied to Pins J1-K, and left, or right limit voltages applied at J3-A and J3-B, the magnitude of the 400 Hz sight azimuth error signal on Pin J1-M shall control the pulse width on the turret left drive Pin J4-D or the turret right drive respectively Pin J4-F. See paragraph 3.6.1.2.14 for current limits. The phase of the signal shall determine whether the left drive output (Pin J4-D) or the right drive output (Pin J4-F) iS enabled. A signal on pin J1-M in phase with the reference voltage on Pin J2-L shall enable the right drive output for a pulse width modulated 28 Vdc (high level) on Pin J4-F and shall disable the left drive output on Pin J4-D. An out of phase signal on Pin J1-M shall enable the left drive output for a pulse width modulated 28 Vdc (high level) on pin J4-D and shall disable the right drive output on Pin J4-F. The pulses of 28 Vdc (high level) peak amplitude Of varying width at pin J4-D or Pin J4-F shall be at a repetitive rate of 600 Hz and shall supply a minimum output load (split series d-c motor) of 0.1 ohm in series with 150 microhenries, +20 percent, Which saturates at 30 amperes. Each output shall have a coasting diode for protection against back electromotive force. The load return shall be on Pin J4-E. The following threshold levels apply:

A voltage on Pin J3-A (azimuth limit-left) of less than 5 Vdc shall disable the left drive output on Pin J4-D. Voltages between 20 and 30 Vdc on Pin J3-A shall enable the output at Pin J4-D.

b. A voltage on Pin J3-B (azimuth limit-right) of less than 5 Vdc shall disable the right drive output on Pin J4-F. Voltages between 20 and 30 Vdc on Pin J3-B shall enable the output on Pin J4-F.

3.6.1 .2.6.2 Elevation output drive and control. With 26.5 (+5) Vdc applied to Pins J1-K, J3-C and J3-K, the magnitude of the 400 Hz signal on Pin J1-B shall determine whether the down drive output (pin J4-A) or the UP drive output (Pin J4-C is enabled. A signal on Pin J1-B in phase with the reference voltage on Pin J2-L shall enable the up drive output for a pulse width modulated 28 Vdc (high level) on Pin J4-C and shall disable the down drive output on Pin J4-A. AN out of phase signal on Pin J1-B shall enable down drive output for a pulse width modulated 28 VdC (high level) On pin J4-A and shall disable the up drive output on Pin J4-C. The pulses of 28 Vdc (high level) at Pin J4-A or Pin J4-C shall be at a repetitive rate of 600 Hz and shall supply a minimum output load of 0.1 ohm in series with 150 microhenries, +20 percent which saturates at 30 amperes. Each output shall have a coasting diode for protection against back electromotive force. The load return shall be on Pin J4-B. The following threshold levels apply:

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a. A voltage on Pin J3-C (elevation down limit) of less than 5 Vdc shall disable the drive ouput on Pin J4-A. Voltages between 20 and 30 Vdc on Pin J3-C shall enable the Pin J4-A output.

b. A voltage on Pin J3-K (elevation Up limit) of less than 5 Vdc shall cause the output current limit on Pin J4-C to be reduced to approximately 20 percent of the normal current limit. Voltages between 20 and 30 Vdc on Pin J3-K shall produce a normal current limit on Pin J4-C.

3.6.1.2.7 Gain characteristics.

3.6.1.2.7.1 Azimuth. The azimuth input rms error signal at Pin J1-M required to produce percent pulse width at Pin J4-F or Pin J4-D shall be 0.0005 to 0.0014 Vrms (above deadband) when the input control voltage at Pin J3-J is 10 Vrms.

3.6.1.2.7.2 Elevation. The elevation input rms error signal at Pin J1-B required to produce a 50 percent pulse width at Pin J4-Aor Pin J4-C shall be 0.0004 to 0.0012 Vrms (above deadband).

3.6.1.2.8 Deadband. The maximum input rms error signal required to form on output pulse at Pins J4-A, C, D or F of the narrowest width and full amplitude shall be as follows:

a. At J1-M (azimuth input) - 10.OmVrms

b. At J1-B (elevation input) - 8.2 mVrms

c. Offset: (1) Azimuth - 1.1 mVrms
(2) Elevation - l.OmVrms

Deadband shall be calculated by dividing the sum of the deadbands measured for each polarity by two. Offset shall be calculated by dividing the difference of the deadband measured for each polarity by two.

3.6.1 .2.9 Drive output waveform characteristics. Waveforms at Pins J4-A, C, D, and F at a 50 percent pulse width shall have a mximum rise and fall time of 20 s. The pulse period shall be 1.67 (+0.33) ms with the pulse width variable from zero percent to 100 percent of the-pulse period. The amplitude of the pulse shall be approximately 2 Vdc less than the 28 Vdc high level at El.

3.6.1.2.10 Velocity limit. The output voltages shall be limited as specified in the following subparagraphs.

3.6.1.2.10.1 Azimuth limit. With 1.1 (+0.1) Vrms applied at Pin J1-M, the drive output shall switch from one output to the other (Pins J4-D or F) when a d-c voltage on Pin J3-R is increased from a low value to 68.75 (+10.25) Vdc (Pin JI-M signal in phase with Pin J2-L reference) or -68.75 (±10.25) Vdc (Pin J1 -M signal out of phase with the Pin J2-L reference voltage).

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3.6.1.2.10.2 Elevation limit. With 1.1 (+0.1) Vrms applied at Pin J1-B, the drive output shall switch from one output to the other (Pins J4-A or C) when a d-c voltage on Pin J3-T is increased from a low voltage to 70 (+11) Vdc or -65 Vdc (+6) Vdc. The polarity required depends on the phase of the voltage on Pin J1-B. For a Pin J1-B signal in phase with the Pin J2-L reference voltage, -65 (+6) Vdc is required at Pin J3-T to switch the outputs. With an out of phase voltage on Pin J1-B, 70 (±11) Vdc is required to switch the outputs.

3.6.1.2.11 Cross talk. When the input to PinJ1-M is such (approximately zero mV) that no pulse width is obtained on that output (Pins J4-D or F) and an input is applied to Pin J1-B to obtain a 50 percent pulse width output on Pins J4-A or C, the other output (Pins J4-D or F) shall continue to produce no pulse width output. The reverse shall occur when the input conditions of Pin J1-B and Pin JI-M are interchanged.

3.6.1.2.12 Gun firing interrupt signals.

3.6.1.2.12.1 Azimuth. With 26.5 (+5) Vdc applied to Pin J1-K, the voltage at pin J2-M (azmuth error gate) shall remain at 28 Vdc (low level) while the input signal at Pin J1-M is increased from zero to a magnitude that produces 1.33 Vdc (+15 percent) at Pin J2-D. With the 1.33 Vdc (+15 percent) at Pin J2-D, the J2-M voltage shall be zero (+0.5) Vdc. When the-voltage at pin J2-D is reduced to 1.09 Vdc (+15 percent), the voltage at pin J2-M shall switch from zero to 28 Vdc (low level).

3.6.1.2.12.2 Elevation. With 26.5 (+5) Vdc applied to Pin J1-K, the voltage at Pin J2-N (elevation error gate) shall remain at 28 vdc (low level) while the input signal at Pin J1-B is increased from zero to a magnitude that produces 1.33 Vdc (+15 percent) at Pin J2-P. With the 1.33 Vdc (+15 percent) at Pin J2-P. The voltage at PinJ2-N shall be zero (+0.5) Vdc. When the voltage at Pin J2-P is reduced to 1.09 Vdc (+15 percent), the voltage at pin J2-N shall switch from zero to 28 Vdc (low level).

3.6.1.2.13 Emergency stow mode and voltage protection. All four drive outputs (Pins J4-A, C, D, and F) shall be disabled when any one of the following conditions occur: (See 4.7.14).

a. The 28 Vdc (high level) at E1 is less than approximately 18 Vdc or greater than 32 to 36 Vdc.

b. The internal -15 Vdc and +15 Vdc power supply voltages or 28 Vdc (low level) current buss are out of tolerance.

c. The d-c voltage source at Pin J3-N is not present.

When any of the above conditions occur (except for the 28 Vdc high level maximum condition), the voltage at Pin J3-N shall be 25.5 to 27.5 Vdc or zero for the condition described in (c) above. Normally, the Pin J3-N voltage shall be 2.5 to 4.0 Vdc with an external test voltage of 28 Vdc applied to Pin J3-N through an external test circuit resistance of 10,000 ohms (±10 percent).

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3.6.1.2.14 Current limiting, drive outputs. When operating with the loads specified in 3.2.1.2.6.1 and 3.2.1.2.6.2, the average current at Pins J4-A, C, D, and F shall be limited in the TCU to 90 (±10) amperes.

3.6.1.2.15 Slow slew velocity limit. When a ground is applied to Pin J2-S, both the azimuth and elevation drive outputs shall be limited as specified in the following subparagraphs.

3.6.1.2.15.1 Azimuth. With 1.1 (+0.1) Vrms applied to Pin J1-M, the drive output shall switch from one output to the other (pins J4-D Or F) when the d-c voltage on Pin J3-R is increased from a Iow value to +4.9 (+0.73) Vdc (the PinJ1-M signal in phase with the PinJ2-L reference) or -4.9 (+0.73) Vdc (the PinJ1-M signal out of phase with the PinJ2-L reference).

3.6.1.2.15.2 Elevation. With 1.1 (+0.1) Vrms applied to Pin J1-B, the drive output shall switch from one output to the other (Pis J4-A or C) when the d-c voltage on Pin J3-T is increased from a low value to +4.9 Vdc +1.5 Vdc or -4.9 Vdc -1.15 Vdc. -0.79 +0.78

3.6.1.2.16 Tachometer dynamic input.

3.6.1 .2.16.1 Azimuth right drive. With a 50 percent pulse width output at Pin J4-F (as specified in 3.2.1.2.7), a voltage rising from zero to 50 Vdc (in approximately 3 seconds) and being applied to Pin J3-R (with Pin J3-P as the return) shall cause the output at J4-F to decrease to zero during the transition. The decrease from +50 Vdc to zero volts at Pin J3-R shall cause the output at Pin J4-F to have a 100 percent pulse width during transition.

3.6.1.2.16.2 Azimuth left drive. With a 50 percent pulse width output at Pin J4-D (as specified in 3.2.l.2.7), a voltage decreasing from zero to -50 vdc (in approximately 3 seconds) and being applied at Pin J3-R (with pin J3-P as the return) shall cause the output at Pin J4-D to decrease to zero during the transition. The increase from -50 Vdc to zero at Pin J3-R shall cause the output at Pin J4-D to have a 100 percent pulse width during the transition.

3.6.1.2.16.3 Elevation up drive. With a 50 percent pulse width output at Pin J4-C (as specified in 3.2.1.2.7) a voltage decreasing from zero to -50 Vdc (in approximately 3 seconds) and being applied at Pin J3-T (with PinJ3-S as the return) shall cause the output at Pin J4-C to decrease to zero during the transition. Increasing from -50 Vdc to zero at Pin J3-T shall cause the output at Pin J4-C to have a 100 percent pulse width during the transition.

3.6.1 .2.16.4 Elevation down drive. With a 50 percent pulse width output at Pin J4-A (as specified in 3.2.1.2.7), a voltage increasing from zero to +50 Vdc (in approximately 3 seconds) and being applied at pin J3-T (with pin J3-S as the return) shall cause the output at Pin J4-A to decrease to zero during the transition. The decrease from +50 Vdc to zero at Pin J3-T shall cause the output at Pin J4-A to have a 100 percent pulse width during the transition.

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3.6.1.2.17 Limit switch dynamic braking. In the following paragraphs, the threshold levels discussed in 3.6.1.2.6.1 and 3.6.1.2.6.2 apply:

3.6.1.2.17.1 Azimuth right drive. Wtih a 50 percent pulse width output at Pin J4-F (as Specified in 3.2.1.2.) and with +20 Vdc on Pin J3-R, remove the power from Pin J3-A. The pulse output on Pin J4-F shall disappear and then reappear in approximately 800 ms.

3.6.1.2.17.2 Azimuth left drive. With a 50 percent pulse width output at Pin J4-D (as specified in 3.2.1.2.1) with -20 Vdc on Pin J3-R, remove the power from Pin J3-B. The pulse output on Pin J4-D shall disappear and then reappear in approximately 800 ms.

3.6.1.2.17.3 Elevation up drive. With a 50 percet pulse width output at Pin J4-C (as specified inn.2.1.2.7) and with -20 Vdc on Pin J3-T, remove the power from Pin J3-C. The pulse at Pin J4-C shall disappear and then reappear in approximately 800 ms.

3.6.1.2.17.4 Elevation down drive. With a 50 percent pulse width output at Pin J4-A (as specified in 3.2.1 .2.7) and with +20 Vdc on Pin J3-T, remove the power from Pin J3-K. The pulse at Pin J4-A shall disappear and then reappear in approximately 800 ms.

3.6.1.3 Elapsed time meter. The TCU shall have a meter that provides a readout in hours of the time that 115 Vac has been applied.

3.6.1.4 Proof test (first article inspection only). Turret Control Units shall operate when assemled to an accepted universal Turret Subsystem (UTS).

3.7 Design and construction.

3.7.1 Production drawings. The TCU shall be fabricated and assembled in accordance with the associated drawings, parts lists and other documents listed on Drawing 11830908.

3.7.2 Standards of manufacture. The TCU shall be manufactured to meet the requirements of the applicable drawings.

3.8 Environmental requirements.

3.8.1 Temperature. The operating temperature of the TCU shall be from -65°F to +125°F and the non-operating temperature shall be from -650F to +160F.

3.8.2 Altitude. The operating altitude of the TCU shall be from sea level to 50,000 feet.

3.8.3 Temperature - altitude combination. The TCU shall meet the requirements of MIL-E-5400, paragraph 3.2.24.3.

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3.8.4 Humidity. The TCU shall withstand the effects of humidity up to 100 percent,including conditions wherein condensation takes place in and on the TCU. The TCU shall withstand the above conditions during operating and non-operating conditions.

3.8.5 Vibration. The TCU shall meet the requirements of MIL-E-5400, paragraph 3.2.24.5.1.3 except as modified by drawing.

3.8.6 Shock. The TCU shall meet the requirements of MIL-E-5400, paragraph 3.2.24.6.

3.8.7 Sand and dust. The TCU shall withstand, in both an operating and non-operating condition, exposure to sand and dust particles as encountered in areas of the world.

3.8.8 Fungus. The TCU shall withstand, in both an operating and non-operating condition exposure to fungus growth as encountered in tropical climates.

3.8.9 Salt fog. The TCU shall withstand, in both an operating and non-operating condition, exposure to salt-sea atmospheres.

3.9 Workmanship. Workmanship and finish shall be in accordance with Mil-STD-434, Requirement 9. Finished items and parts shall not exhibit faulty material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, foreign matter, deformations, and missing operations which may affect serviceability, operation, or safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are hereby made a part of this detail specification.

4.1.1 Certification provisions. The certification provisions of MIL-W-63150 shall apply.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are In accordance with MIL-STD-109.

4.3 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

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4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Turret Control Assembly	11830908	5
Component Assembly	132D1354	5
Component Assembly, Driver	139C7957	5
Case, Electrical Equipment	189F81 9	5
Component Assembly, Demodulator Power Supply	189F960	5
Component Assembly, Interconnection	11830799	5
Component Assembly, Current Loop	11830800	5
Component Assembly Filter Limit Switch	11830909	5

4.4.2 Inspections to be performed. See Table V specified herein.

4.4.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample that fails to comply wih any of the stated requirements.

4.4.4 Electrical interface. Verify that the TCU meets the specified requirements of electrical interface and associated tables by measurement at the pins specified in Table II.

4.4.5 Materials. Verify that materials used are in accordance with the applicable drawings and specifications.

4.4.6 Assembly. Verify that the assembly of the TCU is in accordance with the applicable drawings and specifications.

4.4.7 Interchangeability. Verify that interchangeability of parts is in accordance with MIL-STD-454, Requirement 7.

4.4.8 Performance characteristics. Verify by measuremetn that the TCU meets the performance requirements as stated in 3.6.1.

4.4.9 Electronic characteristics. Verify by measurement that the TCU meets the electronic requirements of 3.6.1.1.

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4.4.10 Environmental tests (First article only).

4.4.10.1 Temperature. A high temperature test shall be performed in accordance with MIL-STD-810, Method 501.1, Procedure II and a low temperature test shall be performed in accordance with MIL-STD-810, Method 502.1, Procedure I. The minimum temperature during the low temperature test shall be -650F, and the maximum temperature during the high temperature test shall be +1600F.

4.4.10.2 Altitude. A low pressure altitude test shall be performed in accordance with MIL-STD-810, Method 500.1, Procedure I.

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TABLE V. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Control Assembly, Turret and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Turret Control Assembly (DWG T1830908)				
	Examination for defects	5		3.3	4.5.4.1
	Environmental	3		3.8	4.4.10
	Interchangeability	3		3.4	4.7
	Proof Test	3		3.6.1.4	4.6.2.2
	Component Assembly (DWG T32D1354)				
	Examination for defects	5		3.3	4.5.4.2
	Component Assembly Driver (DWG T39C7957)				
	Examination for defects	5		3.3	4.5.4.3
	Case, Electrical Equipment (DWG T89F819)				
	Examination for defects	5		3.3	4.5.4.4
	Component Assembly, Demodulator Power Supply (DWG T89F960)				
	Examination for defects	5		3.3	4.5.4.5

NOTES:

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TABLE V. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63534(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Control Assembly, Turret and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Component Assembly, Interconnection (DWG 11830799)				
	Examination for defects	5		3.3	4.5.4.6
	Component Assembly, Current Loop (DWG 11830800)				
	Examination for defects	5		3.3	4.5.4.7
	Component Assembly Filter, Limit Switch (DWG 11830909)				
	Examination for defects	5		3.3	4.5.4.8

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4.4.10.3 Temperature - altitude combination. A temperature - altitude test shall be PerFormed in accordance with MIL-STD-810, Method 504.1. Procedure I and equipment category 5.

4.4.10.4 Humidity. A humidity test shall be performed in accordance with MIL-STD-810, Method 507.1 Procedure I.

4.4.10.5 Vibration. A vibration test shall be performed in accordance with MIL-STD-514.2, Method 514.2, Procedure I Equipment Category C, except use 0.3 inch double AMplitude from 5 to 40 cps or +3 g acceleration, whichever is less, and use 0.036 inch double amplitude from 40 cps to 73 cps and +10 g from 73 Cps to 500 Cps.

4.4.10.6 Shock. A shock test shall be performed in accordance with MIL-STD-810, -516.2, Procedure I.

4.4.10.7 Sand and dust. A dust test shall be performed in accordance with MIL-STD-810, Method 510.1, Procedure I.

4.4.10.8 Fungus. A fungus test shall be performed in accordance with MIL-STD-810, Method 508.1, Procedure I.

4.4.10.9 Salt spray test. Samples shall be selected from each protective finished lot and subjected to the salt spray test in accorcarce with the procedures specified in ASTM B 117. Unless otherwise specified on the applicable drawing the minimum exposure time shall be one and one half (1 1/2) hours using a 5% sodium chloride solution. Rejection shall be in accordance with the applicable protective finish specification.

4.5 Quality conformance (acceptance) inspections. Acceptance inspection shall be performed and shall be in accordance with the following:

4.5.1 Sampling for inspection. Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Contractor in the determination of the acceptability of products shall be in accordance with the provisions set forth in MIL-STD-105, inspection Level II.

4.5.2 In-process control. The contractor shall establish in-process inspection at strategically located points throughout his manufacturing processes to assure continuous control of product quality. In addition, the contractor shall provide and maintain work gages and other measuring and testing devices necessary to accomplish inspection and control quality during his manufacturing processes.

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4.5.3 Inspection records. The contractor shall make available to the Government representative records of qualification of operators and equipment for special processes to applicable specifications and standards. This shall apply whether the special processes are performed at the subcontractor's facility or at the contractor's plant. Inspection records shall be maintained on file at the contractor's plant and made readily available to the Government representative, upon request, for the duration of the contract. The contractor's records of inspection shall provide, as a minimum, the following information:

a. Contract order number.

b. Drawing and specification number with revision letter and revision date.

c. Nomenclature of items inspected.

d. Number of pieces inspected.

e. Individual characteristics inspected.

f. Conformance/nonconformance criteria.

g. Number of pieces accepted/rejected.

h. Description and quantity of defects found; and corrective action taken.

i. Results of inspection (e.g., identification of conforming and non-conforming product). Actual inspectio and test result basis for conformance/nonconformnce decisions.

j. Date of inspection.

4.5.4 Examination. The contractor shall perform inspection for acceptance upon completion of all manufacturing and in-process inspection operations, prior to preservation and packing. Inspection for acceptance shall include, but not be limited to, inspection for workmanship, protective finish, missing features (missing operations), characteristics not previously inspected, and characteristics that may have been altered or changed as a result of subsequent operations after prior inspections were completed.

4.5.5 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.5.6 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" Column in the tables starting with paragraph 4.5.4.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 herein.

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4.6 Inspection methods and procedures. Any unit which fails to comply with the specified requirements or evidences degradation of its operational performance shall be classed defective. Test equipment and procedures shall be satisfactory to the Contracting Officer (see 4.5.6).

4.6.1 Inspection conditions. Unless otherwise specified in the contract or purchase order, the requirements for the turret control unit, Drawing 11830908, shall be verified using the following test conditions:

a. Power as specified in the requirenmt of power supply.

b. Temperature of 70°F ± 20°F.

Measurements will be made using approved functional inspection and/or standard commercial measuring equipment (see 4.5.6).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING N
4.5.4.1	Turret Control Assembly				11830
					NEXT HIGH
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH
Critical	None defined				
Major					
101	Solder workmanship unsatisfactory		100%	3.3	Visua
102	Wiring workmanship unsatisfactory		100%	3.3	Visua
103	Wire marking improper or missing		100%	3.3	Visua
104	Part or assembly damaged		100%	3.3	Visua
105	Part or assembly missing		100%	3.3	Visua
106	Part or assembly incorrectly assembled		100%	3.3	Visua
107	Performance	a	--	3.6	4.6
Minor					
201	Identification plate lettering misleading unidentifiable or missing		1.50%	3.3	Visua
202	Exposed surfaces unprotected		1.50%	3.3	Visua
203	Solder connections not insulated		1.50%	3.3	Visua
204	Poor workmanship		2.50%	3.9	Visua

NOTES:

a: Three samples; accept if one fail, reject if one fails.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.5.4.2	Component Assembly		132D1354
			NEXT HIGHER ASSEMBLY 11830908

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture present on assembly		0.65%	3.3	Visual
102	Moisture protection compound missing		0.65%	3.3	Visual
103	Solder workmanship unsatisfactory		0.65%	3.3	Visual
104	Part missing		0.65%	3.3	Visual
105	Assembly damaged		0.65%	3.3	Visual
Minor					
201	Marking misleading, unidentifiable or missing		1.50%	3.3	Visual
202	Fillet missing		1.50%	3.3	Visual
203	Poor workmanship		2.50%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.4.3	Component Assembly Drive				139C7957
					NEXT HIGHER ASSEMBLY 11830908
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship unsatisfactory		0.65%	3.3	Visual
102	Wire marking improper or missing		0.65%	3.3	Visual
103	Part missing		0.65%	3.3	Visual
104	Assembly damaged		0.65%	3.3	Visual
Minor					
201	Marking misleading, unidentifiable or missing		1.50%	3.3	Visual
202	Poor workmanship		2.50%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 189F819
4.5.4.4	Case, Electrical Equipment				NEXT HIGHER ASSEMBLY 11830908
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Welding workmanship unsatisfactory		0.65%	3.3	Visual
102	Brazing workmanship unsatisfactory		0.65%	3.3	Visual
103	Part missing		0.65%	3.3	Visual
104	Assembly damaged		0.65%	3.3	Visual
Minor					
201	Marking incorrect, unidentifiable or missing		1.50%	3.3	Visual
202	Notched tangs not removed		1.50%	3.3	Visual
203	Poor workmanship		2.50%	3.9	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.4.5	Component Assembly, Demodulator Power Supply				189F960
					NEXT HIGHER ASSEMBLY 11830908
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture present on assembly		0.65%	3.3	Visual
102	Moisture protection is missing		0.65%	3.3	Visual
103	Solder workmanship is poor		0.65%	3.3	Visual
104	Part missing		0.65%	3.3	Visual
105	Assembly damaged		0.65%	3.3	Visual
106	Electrical requirements		100%	3.6	4.6
107	Environmental requirements		--	3.3	Certification
Minor					
201	Marking misleading, unidentifiable or missing		1.50%	3.3	Visual
202	Fillet missing		1.50%	3.3	Visual
203	Compound on prohibited areas		1.50%	3.3	Visual
204	Poor workmanship		2.50%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.5.4.6	Component Assembly, Interconnection		11830799
			NEXT HIGHER ASSEMBLY 11830908

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture present on Assembly		0.65%	3.3	Visual
102	Moisture protection missing		0.65%	3.3	Visual
103	Solder workmanship poor		0.65%	3.3	Visual
104	Part missing		0.65%	3.3	Visual
105	Assembly damaged		0.65%	3.3	Visual
106	Part incorrectly assembled (including wiring)		0.65%	3.3	Visual
Minor					
201	Marking misleading, unidentifiable or missing		1.50%	3.3	Visual
202	Compound on prohibited areas		1.50%	3.3	Visual
203	Poor workmanship		2.50%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-T-63534(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11830800
4.5.4.7	Component Assembly, Current Loop				NEXT HIGHER ASSEMBLY 11830908
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture present on assembly		0.65%	3.3	Visual
102	Moisture protection missing		0.65%	3.3	Visual
103	Solder workmanship poor		0.65%	3.3	Visual
104	Part missing		0.65%	3.3	Visual
105	Assembly damaged		0.65%	3.3	Visual
106	Electrical requirements		100%	3.6	4.6
107	Environmental requirements		--	3.3	Certification
Minor					
201	Marking misleading, unidentifiable or missing		1.50%	3.3	Visual
202	Fillet missing		1.50%	3.3	Visual
203	Compound on prohibited areas		1.50%	3.3	Visual
204	Poor workmanship		2.50%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-T-63534(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11830909
4.5.4.8	Component Assembly, Filer, Limit Switch				NEXT HIGHER ASSEMBLY 11830908
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture present on assembly		0.65%	3.3	Visual
102	Moisture protection missing		0.65%	3.3	Visual
103	Solder workmanship poor		0.65%	3.9	Visual
104	Part missing		0.65%	3.3	Visual
105	Assembly damaged		0.65%	3.3	Visual
Minor					
201	Marking misleading, unidentifiable or missing		1.50%	3.3	Visual
202	Compound on prohibited areas		1.50%	3.3	Visual
203	Poor workmanship		2.50%	3.9	Visual

NOTES:

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Note: Details such as inputs/outputs, signal levels, pin connections, etc., for the tests specified below are in the referenced Section 3 paragraph. This information is a pertinent part of each test specified.

4.6.2 Azimuth preamplifier and demodulator-Action Mode. Verify that the azimuth demodulator output voltage (in the action mode) meets the specified requirements of azimuth preamplifier and demodulator-Action mode and associated table.

4.6.3 Elevation preamplifier and demodulator-Action Mode. Verify that the elevation demodulator output voltage (in the action mode) meets the specified requirements of Elevation preamplifier and demodulator-action mode.

4.6.4 Azimuth preamplifier and demodulator-Fixed Mode/Adjustment. Verify that the azimuth demodulator output (in the fixed mode) can be adjusted and meets the specified requirements of azimuth preamplifier and demodulator-Fixed Mode/Adjustment.

4.6.5 Elevation preamplifier and demodulator-Fixed Mode/Adjustments. Verify that the elevatlon demodulator output (In the fixed mode) can be adjusted and meets the specified requirements of Elevation preamplifier and demodulator-Fixed Mode/Adjustments.

4.6.6 Normal stow mode. Verify that the voltage specified in 3.6.1.2.3 meets the specified requirements of normal stow mode.

4.6.7 Azimuth output drive and control. Verify that the azimuth output voltage meets the amplitude, width and repetitive rate of the specified requirement for azimuth output drive and control and that the azimuth limit, right and left signals disable their respective drive outputs.

4.6.8 Elevation output drive and control. Verify that the elevation output voltage meets the amplitude, width and repetitive rate of the specified requirement for elevation output drive and control and that the elevation limit, right and left signals disable their respective drive outputs.

4.6.9 Gain characteristics. Verify that the azimuth and elevation output gains meet the specified requirements of azimuth gain characteristics and elevation gain characteristics (as related to deadband signals).

4.6.10 Deadband. Verify that the deadband meets the specified requirements of deadband.

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4.6.11 Drive output waveform characteristics. Verify that the azimuth and elevation waveforms meet the specified requirements of Drive output

4.6.12 Velocity limit. Verify that the azimuth and elevation drive outputs change meet the specified requirements of azimuth velocity limit and evaluation velocity limit with application of the velocity limit.

4.6.13 Cross talk. Verify that there is no interaction between the azimuth and elevation outputs as indicated in the specified requirements of cross talk.

4.6.14 Gun firing interrupt signals. Verify that the gun interupt modes meet the speed requirements of azimuth gun firing interrupt signals and elevation gun firing interrupt signals.

4.6.15 Emergency stow mode and voltage protection. Verify that the emergency stow operation mode meets the specified requirements of emergency stow mode and voltage protection.

4.6.16 Current limiting, drive outputs. Verify that the current limit for each drive output meet the specified requirement of current limiting, drive outputs.

4.6.17 Slow slew velocity limit. Verify that the limits for azimuth and elevation slew meet the specified requirements of slow slew velocity limit.

4.6.18 Tachometer dynamic input. Verify that the limits for the tachometer inputs meet the llmits Of the specified requirement of tachometer dynamic input.

4.6.19 Limit switch dynamic braking. Verify that the limits meet the specified requirements of limit switch dynamic braking.

4.6.20 Elapsed time meter. Verify that the elapsed time meter advances one hour when a-c power to the gun control has bee nactivated for one hour.

4.6.2.1 Workmanship. The specified requirements of assembly shall be verified by continuous inspection of all parts and assemblies to assure conformance to high quality workmanship.

4.6.22 Proof test (first article inspection only). Uhen specified in the contract (see 6.2) three 3) Turret Control Units shall be forwarded to the site specified by the contracting agency and assembled to an accepted Universal Turret Subsystem (UTS). The Turret Control Units shall function in accordance to the intent of the design and with no evidence of malfunction. When approved by the contracting agency, this test may be performed at the aircraft system level.

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4,7 Interchangeability testing. At unit which fails to comply with the specified requriement or evidences degradation of its operational performance shall be classed defective. Test equipment and procedures shall be satisfactory to the contracting officer (see 4.5.6).

4.7.1 Turret control assembly. A sample of three turret control assemblies randomly selected, shall be tested for interchangeability using the test method specified in 4.7.3. Control assemblies taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. Three control assemblies shall comply with the applicable specification functioning requirements of 4.4.10, 4.5 and 4.6 before and after the interchange of parts. It shall be the responsibility of the agency responsible for the test to inform the contracting officer by letter or other methods satisfactory to the contracting officer, when selective assembly has become necessary. Statement shall contain, as a minimum, contractor's name and FSCM, contract number, parts involved in the selective fit, conditions of parts and rationale for action taken. No hand refinement of parts shall be permissible without approval by the contracting officer. No malfunctions shall be allowed in the functioning tests. Failure of a control assembly in the interchangeability test shall cause retest or rejection of the represented lot.

At the discretion of the Government representative, an interchangeability retest may be allowed without reconditioning the lot. Failure during the retest shall cause rejection of the represented lot subject to reconditioning and further test as a reconditioned lot. A sample of six control assemblies from each retest or reconditioned lot shall be tested using the same procedures described above.

4.7.2 Repair parts. At least two parts from each inspection lot of concurrent repair parts shall be subjected to the interchangeability test specified i 4.6.3. Failure of any part to meet the interchangeability requirements shall be cause for rejection of the represented lot, subject to reconditioning and further test as a reconditioned lot. A sample consisting of "double the number of parts used in the original test shall be tested from each reconditioned lot using the test methods specified above.

4.7.3 Test method for interchangeability. Control assemblies shall be tested by disassembling and then reassembling the parts specified below. Parts shall be disassembled from the control and identical parts placed together and mixed. Any commercial part (screws, pins, etc.) rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test.

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List of Interchangeable Parts

Component	Drawing
Component Assembly	132D1 354
Component Assembly Driver	139C7957
Case, Electrical Equipment	189F81 9
Component Assembly, Demodulator Power Supply	189F960
Component Assembly, interconnection	11830799
Component Assembly, Current Loop	11830800
Component Assembly, Filter Limit Switch	11830909

4.7.4 Repair parts. Concurrent repair parts shall be tested by disassembling two control assemblies previously tested in 4.7.3 and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed and the assembled control assemblies shall operate and function as intended. This test may be performed independently of the control assembly interchangeability testing specified in 4.7.1 and at more frequent intervals using accepted drive assemblies taken from current production.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Controls shall be preserved, packaged, packed and marked in accordance with the requirements of the Preservation and Packaging Data for Drawing P 11830908 at the level of protection specified in the contract (See 6.2b).

5.1.1 Spare parts or replacement parts. Spare parts Br replace-fit parts shall be cleaned, preserved, packaged and packed as specified i nthe contract or order.

6. NOTES

6.1 Intended use. The TCU is to be used to control the XM97E1 20mm Armament subsystem turrent subassembly.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Packing and shipping instructions.

c. Government furnished property (if required).

d. Preproduction sample and test.

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6.2.2 Contract data requirements. Contract data requirements for inspection equipment design sall conform to Data Item Description DI-R-1714, Tailored.

6.3 Submission of inspection equipment design for approval. See MIL-W-63150. Equipment designs shall be submitted as required to: Commander, ARRADCOM, ATTN: DRDAR-QAF-1, Dover, NJ 07801. Request letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects and test paragraphs.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otberwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commander, ARRADCOM, ATTN: DRDAR-QAF-S, Dover, NJ 07801.

Custodian:
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(Project 1005-A615)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-63534 (AR) TURRET SUBSYSTEM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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OF VALIDATION

MIL-T-63534(AR)
NOTICE 1
22 MARCH 1988

MILITARY SPECIFICATION

TURRET SUBSYSTEM UNIVERSAL (UTS), FOR ARMAMENT SUBSYSTEM, HELICOPTER, 20mm: M97E1 AND M97E2, TURRET CONTROL UNIT FOR

MIL-T-63534(AR), dated 27 December 1981, has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Army - AR

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